/backend

Execution backend surface: /backend [] | /backend show | /backend help

Overview

`/backend` is the TUI surface for the execution backend (spec `openhands` FR-008). An execution backend selects **where** a session's tool invocations actually run: `local` runs tools on the host exactly as before (the default), `docker`/`podman` run them inside a sandbox container, and `remote` drives a second ragent server over its REST+SSE API.

With no argument the command opens an interactive **switcher panel** that shows the active backend and the health of *every* registered backend. Each row carries the backend's id, kind, health (`ok` / `unavailable` / `unknown`), and a secret-free connection summary. The panel is keyboard-driven: `Up`/`Down` move the block cursor, `Enter` switches the active backend to the highlighted row, and `Esc` closes the panel.

The panel is a modal overlay: while it is open it swallows every keystroke and the message input field stays locked.

Syntax

```
/backend                # open the switcher panel
/backend show           # alias of /backend
/backend list           # alias of /backend
/backend local          # switch to the local host backend
/backend docker         # switch to a Docker sandbox backend
/backend podman         # switch to a Podman sandbox backend
/backend remote         # switch to a remote backend
/backend <id>           # switch to a backend by its registered id
/backend help           # usage help
```

Options / Subcommands

Form	Description
`/backend`	Open the switcher panel listing the active backend and every registered backend's health
`/backend show` / `/backend list`	Alias of `/backend`
`/backend <kind>`	Switch directly to a backend kind (`local`, `docker`, `podman`, `remote`)
`/backend <id>`	Switch directly to a backend by its registered id (from `backends` in `ragent.json`)
`/backend help`	Print the usage table (no state change)

Panel keys

Key	Action
`Up` / `Down`	Move the block cursor between rows

Key	Action
`Enter`	Switch the active backend to the highlighted row
`Esc`	Close the panel

A row whose health is not `ok` is dimmed and **cannot** be selected: the switch is refused and the active backend is left unchanged, so the switcher can never leave the session on a backend that cannot run a tool.

Health

Kind	`ok` when	`unavailable` when
`local`	always (the host is always available)	never
`docker` / `podman`	the runtime binary resolves on `PATH` (FR-026)	the binary is missing (the detail names it)
`remote`	a base URL and a key are configured (FR-030)	either is absent

Health is a snapshot taken when the panel opens; the panel does not re-probe on each frame.

Persistence

A successful switch writes `execution_backend: "<kind>"` into the loaded config file (the project-local `.ragent/ragent.json` when one was loaded, otherwise the global config). The per-turn config cache is invalidated so the next turn resolves and dispatches through the new backend.

A descriptor-free kind switch (`/backend docker`) writes the bare label. To carry an image, workspace, URL, or key, declare a full descriptor in the `backends` array (or as an inline `execution_backend` object) and select it with `/backend <id>`.

Examples

Open the switcher:

```
/backend
```

Switch to a declared Docker sandbox by id:

```
/backend docker-box
```

Return to the host:

```
/backend local
```

Output

- The panel renders a bordered modal titled `Execution backend - active: <kind>`; the active row carries an `[active]` marker.
- A switch appends an `From: /backend` message bubble and sets the status to `backend: <kind>`.

- A refusal appends a `[warn]` message explaining why and sets the status to `backend: unavailable` / `backend: unknown`.
- No permission prompt is raised and no LLM call is made.

Related

- `/backend help` - usage table
- `docs/howtos/sandbox-backends.md` - full guide to the switchable backends, the sandbox mechanism, and a worked Docker/Podman example
- `specs/openhands/SPEC.md` - the execution backend design (`local`/`docker`/`podman`/`remote`)
- `docs/howtos/config.md` §7.36 - the `execution_backend` and `backends` config keys